Benchmark results

NewTOS on Raspberry PI 4

unknown AArch64 TOS 4.04 1920x1080 256c blit none

115 of 115 tests run, 1.0 s each. Values are operations per second (higher is better). Bars compare tests within each group.

Groups

VDI primitives	38 tests run
AES windows & events	21 tests run
Alignment (blitter)	20 tests run
Full-screen windows	17 tests run
GDOS fonts	14 tests run

VDI primitives

Test	ops/s	Note
Line, 2 points	23404	
Polyline, 32 points	699.5	
Wide line (7px, round)	4494.6	
Dashed line	21152	
User line style	19139	
XOR line	13114	
Markers x16	22967	
Filled rect 32x32	90111	
Filled rect, full window	578.2	
Bar, hatched + border	50954	
Pattern fill rect	91444	
User pattern fill	90601	
XOR filled rect	118723	
Polygon, 8 points	17894	
Filled circle	2414.1	
Filled ellipse	2472.4	
Arc	21870	
Pie slice	2508.8	
Elliptical arc	21977	
Elliptical pie	2981.5	
Rounded box	42504	
Filled rounded box	10333	
Contour fill 64x40	7278.6	
Text, 15 chars	30414	
Text, 84 chars	5388.5	
Text with effects	19050	
Text rotated 90	32282	
Justified text	18962	
Text extent query	396263	
Blit scr->scr 64x64	37925	
Blit scroll window	237.8	
Blit mem->scr 64x64	27835	
Blit scr->mem 64x64	24765	
Mono blit (vrt_cpyfm)	45721	
Transform form 64x64	n/a	not supported here
Get pixel	504122	
Hide/show mouse	20377	
Attribute calls x3	170021	

Open/close workstation	47034	

AES windows & events

Test	ops/s	Note
Window create+delete	135279	
Window open+close	217.5	
Window move	657.0	
Window resize	831.0	
Window top (2 windows)	425.7	
Set window title	1441.3	
Set info line	1441.3	
Set slider pos+size	2026.7	
Scroll window content	222.3	
wind_get work area	338447	
Rectangle list walk	155297	
wind_calc	279536	
wind_update pair	158298	
objc_draw dialog	4131.8	
objc_find	294911	
form_dial redraw 64x48	5768.0	
Menu bar redraw	4829.2	
graf_mkstate	340044	
graf_mouse off+on	19319	
Event poll (evnt_multi)	127874	
Message round trip	58236	

Alignment (blitter)

Test	ops/s	Note
Fill 64x64, aligned	117027	
Fill 64x64, x+3	45935	
Blit 64x64, aligned	53342	
Blit 64x64, src+3 dst+11	36318	
Mem->scr 64x64, aligned	28387	
Mem->scr 64x64, x+3	27436	
Mono blit, aligned	47034	
Mono blit, x+3	44171	
Text, x on 8 px	35963	
Text, x+3	30414	
Scroll up, aligned edges	220.7	
Scroll up, edges x+5	208.4	
Scroll left 16 px	186.3	
Scroll left 3 px	186.3	
Move window, aligned	635.8	work x mod 16 = 0
Move window, off-grid	620.3	work x mod 16 = 5
Open/close win, aligned	224.4	work x mod 16 = 0
Open/close win, off-grid	219.7	work x mod 16 = 5
Scroll window, aligned	783.9	work x mod 16 = 0
Scroll window, off-grid	733.9	work x mod 16 = 5

Full-screen windows

Test	ops/s	Note
Open/close full window	131.7	work 1896x1018
Full <-> normal size	235.4	
Top full / test window	273.3	work 1896x1018
Fill full window	377.3	work 1896x1018
Scroll up 1 line	176.0	work 1896x1018
Scroll left 16 px	174.1	work 1896x1018
Scroll left 8 px	127.6	work 1896x1018
Scroll left 3 px	122.7	work 1896x1018
Scroll text 8pt	169.0	system font
Scroll text 10pt	173.2	system font
Scroll text 12pt	173.2	got 10pt, system font
Scroll text 14pt	173.2	got 10pt, system font
Scroll text 18pt	171.5	system font
Scroll text 24pt	180.1	got 20pt, system font
Scroll text 36pt	180.1	got 20pt, system font
Scroll text 48pt	180.1	got 20pt, system font
Scroll, size per line	175.0	system font

GDOS fonts

Test	ops/s	Note
Text 6pt	47719	got 8pt, system font
Text 8pt	47952	system font
Text 10pt	23183	system font
Text 12pt	23075	got 10pt, system font
Text 14pt	22967	got 10pt, system font
Text 18pt	16220	system font
Text 24pt	8612.1	got 20pt, system font
Text 36pt	8454.1	got 20pt, system font
Text 48pt	8776.1	got 20pt, system font
Text 72pt	8532.4	got 20pt, system font
Text 10pt bold	22303	system font
vst_point, 8 sizes	398114	
Font switch + vst_point	n/a	not supported here
Arbitrary size + text	n/a	not supported here
Scroll, mixed sizes	202.9	system font
Text extent at 12pt	396263	
Enumerate font names	n/a	not supported here
Load+unload fonts (x3)	n/a	not supported here

Configuration

Machine

```
Description     : NewTOS on Raspberry PI 4
Machine type    : unknown
CPU             : AArch64 (native)
FPU             : none reported
Blitter         : not present
ST RAM          : 14336 KB total
TT/Fast RAM     : 729088 KB total
```

Operating system

```
TOS             : 4.04  (11/08/2026 m/d/y)
Country / video : USA, NTSC
GEMDOS          : 0.48
MiNT            : not running
Largest free    : ST 10240 KB
                  TT/Fast 715424 KB
```

AES / environment

```
AES version     : 3.40
Multitasking    : no (max apps 1)
Theme pens      : green 166, red 122, blue 149, cyan 155, highlight 106
Application id  : 0
appl_getinfo    : not available
Desktop area    : 0,19 1920x1061
AES char / box  : 8x16 / 19x19
Current dir     : C:\APPS\FBENCH
PATH            : /dev/c/
```

Display

```
Resolution      : 1920x1080, Getrez 4 (TT Medium)
Colours         : 256 on screen, 16 planes
Palette         : 0 (direct colour)
Lookup table    : no (true colour)
Pixel size      : 278 x 278 microns
Screen memory   : 4147200 bytes at $3dbf5000
VDI handles     : physical 1, ui 3 (bench ws opened only for runs)
VDI max vertices: 1024
Text rotation   : 90 degree steps
Writing modes   : 4
```

GDOS / fonts

```
GDOS            : none (system font only)
Fonts           : 1 (0 loaded by GDOS)
  > id     1  6x6 system font
```

Cookie jar

```
NTOS _MCH _IDT _AKP EdDI
```